

ZAHEER AHAMAD MOHAMMED

Mechanical Engineer — CAD Design & Product Development

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Mechanical Engineering graduate with hands-on experience in CAD design, product development, SAP MM operations, robotics, embedded systems, and computer vision. Skilled in SolidWorks, NX CAD, ANSYS, OpenCV, ESP32/Arduino, and industrial workflow practices.

EXPERIENCE

NPD Intern — Pricol Precision Products Pvt. Ltd.

Feb 2026 – May 2026

- Supported NPD activities for automotive components — design documentation, production follow-ups, and quality processes across departments.
- Managed 20+ daily SAP MM transactions (PR creation, Service Entry, R&R, RGP, payment requests) for material and vendor operations.
- Reviewed CMM inspection reports and maintained engineering drawings/dispatch documentation to support quality validation and traceability.

Intern — IIT Tirupati

May 2025 – Jun 2025

- Developed a cam-driven ornithopter mechanism (BLDC-powered) and optimized flapping-wing geometry through simulation-driven aerodynamic analysis.

SKILLS

CAD / Design	NX CAD, SolidWorks, AutoCAD, Fusion 360, ANSYS, Creo
SAP / Industrial	SAP MM, PR Processing, Service Entry, R&R, RGP, Documentation
Programming	Python, Embedded C, ESP8266, ESP32, Arduino
AI & Vision	OpenCV, YOLO, MediaPipe
Core Areas	NPD, Motion Simulation, Robotics, GD&T, Quality Control

PROJECTS

CAD Assembly Design & Motion Simulation (NX CAD): Modeled 10+ multi-component mechanical assemblies (Pipe Vice, 4-Cylinder Engine, Bench Vice, Knuckle Joint, Wheel Support) with GD&T, BOM, and NX Motion Simulation to validate functional movement and clearances.

Face Detection & Recognition Drone (DJI Tello): Built a real-time aerial face detection/tracking system with OpenCV and Tello SDK for perimeter monitoring.

Humanoid Robot (ESP8266): Servo-based head/multi-directional movement with Wi-Fi control and AI voice/text interaction.

5-DOF Robotic Arm (ESP32 + Arduino): Dual control modes — ESP32 Wi-Fi and OpenCV + MediaPipe gesture control — with optimized PWM servo actuation.

EDUCATION

B.Tech, Mechanical Engineering — JNTUA Narayana Engineering College, Nellore

2022 – 2026

CGPA Equivalent: 80%

Intermediate — Narayana Junior College

2020 – 2022

CERTIFICATIONS

- Master Diploma in Mechanical CAD (195 Hrs) — CAD Desk; ANSYS Workbench, Creo, GD&T, NX CAD — Grade: Excellent
- SolidWorks & AutoCAD Essentials Certifications — SourceCAD
- Virtual Internship, Conceptual CAE Design & Simulation — EduSkills | Robotics Certification — Sanfoundry

ACHIEVEMENTS

- 40-hour AI Bootcamp at IIT Hyderabad (Techobyte) — Letter of Recommendation, Appreciation & Internship Offer.
- 1st Prize — Yantra 2K23, 2K24 & 2K25 Project Expos (drone innovation); 1st Prize — district-level gesture-controlled drone competition.
- Grade A (94%) — Sanfoundry Robotics Certification Test.